

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459413.7285 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.048
Transit depth (Rp/Rs)^2	3.24 +/- 1.74 [%]
Orbital Inclination (inc)	89.44 +/- 1.7
Airmass coefficient 1 (a1)	1.065 +/- 0.045
Airmass coefficient 2 (a2)	-0.032 +/- 0.025
Scatter in the residuals of the lightcurve fit is	4.29 %
Best Comparison Star	#3 - [554, 225]
Optimal Aperture	2.38
Optimal Annulus	8.98
Transit Duration (day)	0.126 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.048 vs 0.1326 (deviation 0.95σ)
Duration fit vs published	0.126 d vs 0.1238 d (deviation 0.2σ)
Mid-time O–C	-13.2 min
Depth SNR	3.6
Residual scatter	4.29 %

Light curve

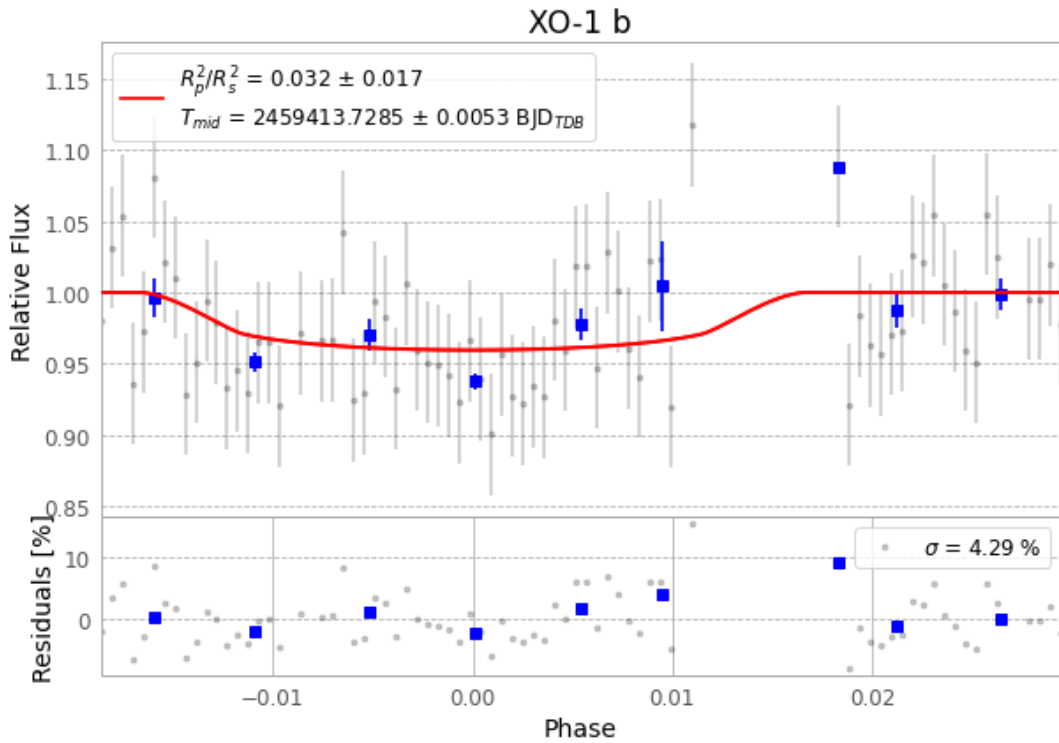


Figure 1 — FinalLightCurve_XO-1 b_2021-07-17.png (SHA-256 6715054650d8bfa3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d057fa5bd3139133...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), XO-1210718033913.FITS → XO-1210718081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-210718.log (c0782a0ba543...), FinalLightCurve_XO-1 b_2021-07-17.png (6715054650d8...), FinalLightCurve_XO-1 b_2021-07-17.csv (85f995b26cbc...)

Compute resources

Wall time 190.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:26Z.